

NBA Player Performance Analysis

Instructors Dr. Rami Jomaa
TA. Zainab Abdulhadi

Group Members:

- Juri Almarwani 4410761
- Nejoud Fahad Alharbi 4410570
- Shahad Tahir 4410656

Problem Statement

Include:

- Modern basketball relies heavily on data analytics
- Traditional analysis cannot capture complex performance patterns
- Machine learning enables predictive and descriptive insights
- NBA player performance is highly nonlinear and context-dependent

How can machine learning help understand and predict NBA player performance?

Project Objectives

- Predict player scoring performance using regression models
- Identify player archetypes using clustering techniques
- Predict game outcomes (Win/Loss) using classification models
- Analyze how player performance changes over time (bonus task)

Project Structure

This project consists of three core machine learning tasks and a bonus task:

- Regression — Player performance prediction
- Clustering — Player style identification
- Classification — Game outcome prediction
- Temporal Analysis — Performance trends over time

Dataset Overview

- Dataset: NBA Player Game Statistics (2024/2025 season)
- Source: Kaggle
- Each row represents one player's performance in one game
- Suitable for prediction, clustering, and time-series analysis

Key Dataset Features

Scoring & Efficiency

- PTS, FG, FGA, FG%
- 3P, 3PA, 3P%
- FT, FTA, FT%

Playmaking & Defense

- AST, STL, BLK, TRB, TOV

Context

- Minutes Played (MP)
- Game Result (Win/Loss)

Data Cleaning & Preprocessing

- Removed games where players recorded 0 minutes
- Handled missing values in shooting percentages
- Retained outliers to preserve exceptional performances
- Normalized numerical features for scale-sensitive models

Feature Engineering

To improve model performance and interpretability, we created new features:

- True Shooting Percentage (TS%)
- Assist-to-Turnover Ratio
- Defensive Impact Score
- Rolling Averages (Temporal trends)

These features capture efficiency, decision-making, and defensive contribution.

TASK 1 — REGRESSION

Objective:

Predict the number of points (PTS) a player scores in a game.

Why regression?

- PTS is a continuous numerical variable
- Useful for performance forecasting and analytics

Regression Models Used

We compared three regression models:

- **Linear Regression**
 - Simple, interpretable baseline
- **Random Forest Regressor**
 - Handles nonlinear relationships
 - Captures feature interactions
- **Support Vector Regression (SVR)**
 - Effective for complex decision boundaries

Models were evaluated using:

- RMSE (Root Mean Squared Error)
- R^2 Score (Explained Variance)

These metrics measure prediction accuracy and model reliability.

Regression Evaluation Metrics

Key Findings:

- Linear Regression struggled with nonlinear patterns
- SVR performed well but required tuning
- Random Forest achieved the lowest RMSE and highest R^2

Conclusion:

Random Forest Regressor is the most reliable model for predicting player points.

Regression Results

TASK 2 — CLUSTERING

Objective:

Identify natural groupings of players based on playing style and statistics.

Why clustering?

- No predefined labels
- Helps discover hidden player archetypes

Clustering Methods

We applied:

- K-Means (Required)
- Gaussian Mixture Models (GMM)

Additional techniques:

- Feature scaling
- Elbow Method & Silhouette Score
- PCA for visualization

Player Archetypes Identified

Three main player groups emerged:

1. High-Usage Players

- High scoring and offensive responsibility

2. Balanced Contributors

- Moderate scoring, rebounding, and efficiency

3. Role Players

- Lower usage, defensive or specialized roles

GMM allowed overlapping roles, reflecting real basketball dynamics.

TASK 3 — CLASSIFICATION

Objective:

Predict whether a team wins or loses based on player performance statistics.

Target Variable:

- Game Result (Win / Loss)

Classification Models Used

We trained and compared:

- Logistic Regression (baseline)
- Random Forest Classifier
- Support Vector Machine (SVM)

Hyperparameter tuning was applied to improve performance.

To avoid biased predictions:

- Used balanced class weights
- Applied stratified train-test split
- Evaluated using multiple metrics beyond accuracy

Handling Class Imbalance

Classification Results

Metrics used:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Best Model:

Random Forest Classifier —
highest overall performance and
stability.

BONUS TASK

Temporal Performance Analysis

Goal:

Analyze how player performance evolves during the season.

Methods:

- Rolling averages (5-game windows)
- Trend visualization

Insights:

- Performance cycles
- Momentum and fatigue effects

- Random Forest performed best in regression and classification
- GMM provided more flexible and realistic clustering
- Ensemble methods handled complexity effectively

Model Comparison Summary

Error Analysis & Limitations

Sources of error:

- Injuries and rest not included
- Opponent strength missing
- In-game context unavailable

These factors reflect real-world unpredictability.

Conclusion

- Machine learning provides strong predictive and descriptive power
- Integrated approach yields deeper insights
- Results are interpretable and practically meaningful

Future Work

- LSTM-based time-series models
- Player tracking and positional data
- Advanced clustering techniques

Team Contributions

- **Juri:** Methodology documentation, classification modeling, hyperparameter tuning, co-authoring the full written report
- **Nejoud:** Distribution analysis, feature engineering, clustering analysis, bonus task, co-authoring the full written report
- **Shahad:** Data cleaning, descriptive statistics, handling missing values, regression modeling, final presentation preparation
- **All Members:** Reviewed final work and ensured overall project completeness and quality

Thank You